

Technology Stack

Living inventory of technologies used in BankOne. **Update this document whenever dependencies, runtimes, or tooling change** (add, upgrade, remove, or replace). Source of truth for versions:

`BankOne-BackEnd/pom.xml` and `BankOne-Frontend/package.json`.

Last reviewed: 2026-07-25

At a glance

Layer	Primary stack
Backend Frontend Database Messaging Email Auth Deploy (primary) Deploy (cloud) Docs tooling	Java 21 · Spring Boot 4.1 · Maven · WAR Angular 22 · TypeScript 6 · npm · SCSS PostgreSQL Apache Kafka (local Docker / Aiven on Render) Mailpit (local SMTP) · Twilio SendGrid HTTPS (Render) Spring Security · JWT (JJWT) Open Liberty (:9080 / :9443 , JDWP :7777) Render (Docker -Pdocker) · Netlify (Angular) Pandoc · Google Chrome (PDF) · python-docx

Backend (`BankOne-BackEnd/`)

Language & build

Technology	Version / notes	Where defined
Java	21	<code>pom.xml</code> <code>java.version</code> <code>.mvn/ pom.xml</code> <code>pom.xml</code> <code><packaging></code>

Spring & persistence

Technology	Purpose
spring-boot-starter- webmvc	Web / REST (<code>@RestController</code>)
spring-boot-starter- data-jpa	JPA repositories, entities, specifications
Hibernate (via Spring Data JPA)	ORM; <code>ddl-auto=update</code> ; SQL logging in <code>application.properties</code>
spring-boot-starter- validation	Bean Validation (<code>@Valid</code> , constraints)
spring-boot-starter- actuator	Actuator endpoints
spring-boot-devtools	Hot reload (runtime, optional)

Security & tokens

Technology	Version / notes
spring-boot-starter-security	Stateless session; filters; method/URL authorization
JJWT (jjwt-api, jjwt-impl, jjwt-jackson)	0.12.7 — HMAC JWT create/parse

Database & JDBC

Technology	Notes
PostgreSQL	Database <code>bankone</code> , JDBC URL in <code>application.properties</code>
PostgreSQL JDBC driver	<code>org.postgresql:postgresql</code> (runtime)
schema.sql / data.sql	SQL init (<code>spring.sql.init.mode=always</code>)

Messaging & email

Technology	Purpose
spring-boot-starter-kafka	Kafka produce/consume (<code>com.bankone.notification</code>)
spring-boot-starter-mail	SMTP path (Mailpit locally)
Apache Kafka 3.9	Local: <code>docker compose</code> service <code>kafka</code> (<code>apache/kafka:3.9.0</code> :9092)
Aiven Kafka	Render managed broker (SASL_SSL + SCRAM + CA PEM)
Mailpit	Local fake inbox SMTP :1025 / UI :8025
Twilio SendGrid	Render email via HTTPS API when <code>MAIL_TRANSPORT=sendgrid</code>

Code generation & container

Technology	Notes
Lombok	Optional; annotation processing in <code>maven-compiler-plugin</code>
spring-boot-starter-tomcat	Embedded Tomcat for <code>spring-boot:run</code> / local WAR (<code>provided</code> for Liberty deploy)

Test dependencies

Technology
spring-boot-starter-data-jpa-test spring-boot-starter-security-test spring-boot-starter-validation-test spring-boot-starter-webmvc-test

Frontend (BankOne-Frontend/)

Core framework

Technology	Version / notes	Where defined
Angular	^22.0.x (core, router, forms, common, animations, compiler, platform-browser) ~6.0.2 ~7.8.0 ^2.3.0 11.16.0 (packageManager) Component / global styles	package.json

UI libraries

Technology	Version / notes
Angular Material	^22.0.5 — buttons, dialogs, forms, icons, cards, etc.
Angular CDK	^22.0.5 — Material peer dependency

Build & quality tooling

Technology	Version / notes
Angular CLI @angular/build Prettier Vitest jsdom	^22.0.7 ^22.0.7 — application builder ^3.8.1 ^4.0.8 — unit tests (ng test) ^28.0.0 — test DOM

Runtime patterns (no extra packages)

Concern	Approach
HTTP Auth	Angular HttpClient JWT in browser storage; Bearer interceptor; authGuard + roles App shell + sidebar under core/layout

Infrastructure & deployment

Technology	Role
Open Liberty	Primary app server (HTTP 9080, HTTPS 9443); scripts/redeploy-liberty.sh
Embedded Tomcat	Alternate local run via Spring Boot (server.port=8080)
PostgreSQL	Persistence (localhost:5432 typical)

Documentation toolchain

Technology	Role
Markdown Pandoc Google Chrome python-docx	Working copies in docs/_source/ Markdown → DOCX / HTML Headless print → PDF Post-process DOCX styling (scripts/_polish_docx.py) Rebuild all DOCX + PDF pairs

IDE / agent conventions (project)

Item	Notes
Cursor rules	<code>.cursor/rules/</code> (docs architect, teach-while-building, deferred hardening) Docs updated with every code change (see <code>documentation-architect.mdc</code>)

How to keep this file current

1. After changing `pom.xml` or `package.json`, update the matching tables above (name + version + purpose).
2. After changing deploy target, DB, or auth mechanism, update **At a glance** and the relevant section.
3. Append an entry to `CHANGELOG.md`.
4. Run `Bank0ne/scripts/regenerate-docs.sh` so DOCX/PDF stay aligned.

Do **not** list planned libraries here until they are actually added to the build.